

Morning Paper Report - Friday, August 14, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Probabilistic Tuning of Attentional Templates: Evidence for Flexible Prioritization in Visual Working Memory

Authors unavailable
2026

Source: OSF/PsyArXiv; matched terms: visual working memory, visual search, working memory, flexibility

Link: https://osf.io/preprints/psyarxiv/yzun9_v4/

Goal: Attentional templates in visual working memory (VWM) are essential for guiding visual search

Method: While much debate has focused on whether these templates are stored in discrete states or continuous resources, a question remains: to what extent can the priority of a template be strategically tuned to match environmental probabilities? We investigated this flexibility by manipulating the reliability of retro-cues (the probability that a cued item is the target) across three experiments combining memory recall...

Results: Together, these results show that the influence of a cued representation varies systematically with cue reliability

Conclusion: Overall, our findings suggest that VWM prioritization is sensitive to expected relevance and adapts to memory load and task structure.

2. A Noisy Guessing Mixture Circular Diffusion Model of Visual Working Memory

Authors unavailable
2026

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory, precision

Link: https://osf.io/preprints/psyarxiv/pg5tk_v1/

Goal: Visual working memory (VWM) is central to human cognition, providing an interface between visual processing and higher-level cognition

Method: While several cognitive models have been proposed to explain VWM, typically assuming that recall error distributions in continuous reproduction tasks reflect a mixture of memory and guessing processes, they suffer from two key limitations

Results: Our simulations show that the NGM-CDM reliably estimates the relative noisiness of guessing and memory processes, demonstrating its potential as a measurement tool

Conclusion: Specifically, these models (i) focus on response precision, while ignoring response times or failing to capture them accurately, and (ii) suggest guessing as a theoretically motivated process while leaving its underlying process unspecified or overconstrained

3. Dynamic prioritization reshapes neural geometries for action in human working memory

Ivan Padezhki, Juan Linde-Domingo, Carlos González-García
2025

Source: Crossref; matched terms: visual working memory, working memory representation, working memory

Link: <https://doi.org/10.1101/2025.07.18.665374>

Goal: Abstract Prior work has shown that control brain regions encode upcoming novel instructed actions

Method: We addressed these questions using EEG while human participants performed two consecutive choice reaction tasks

Results: Abstract Prior work has shown that control brain regions encode upcoming novel instructed actions

Conclusion: A cue then indicated whether these mappings would be relevant immediately ("current" condition), later after an intervening task ("prospective"), or after a free delay ("delayed")

4. Time Regained: Decision Processes in Continuous-Outcome Retrieval from Visual Working Memory

Authors unavailable

2025

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory, precision

Link: https://osf.io/preprints/psyarxiv/ekvgu_v2/

Goal: Decision-making models of memory retrieval hold that across-trial variability in memory representations and within-trial variability in matching the contents of memory to the retrieval cue both contribute to variability in the speed and accuracy of retrieval decisions

Method: Decision-making models of memory retrieval hold that across-trial variability in memory representations and within-trial variability in matching the contents of memory to the retrieval cue both contribute to variability in the speed and accuracy of retrieval decisions

Results: VWM precision followed either an inverse square-root law or power law in set size, depending on saturation, as predicted by sample-size and attention-weighted sample-size models, respectively

Conclusion: The results support a model in which the capacity limits of VWM reflect the statistical properties of a population of noisy neurons that are divided among items in memory under attentional control and determine the precision of recalled items.

5. Young children monitor the fidelity of visual working memory.

Jessica B. Applin, Melissa M. Kibbe

Journal of Experimental Psychology: Learning, Memory, and Cognition, 2021

Source: Crossref; matched terms: visual working memory, working memory, fidelity

Link: <https://doi.org/10.1037/xlm0000971>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.